
GLOBAL CLIMATE SCHEMA : MySQL Database

GLOBAL CLIMATE CHANGE IMPACT & SUSTAINABILITY MONITORING

Database Management Systems — TAE 2

PRESENTED BY

Krushna Panchal

TY. B.Tech CSE (Data Science)

INSTITUTE

G H Raison College of Engineering
and Management, Pune

DATABASE

MySQL WorkBench

Problem & Objective

MOTIVATION



Problem

- I. Climate datasets contain large numbers of yearly observations.
- II. Raw datasets contain missing values and inconsistent formats.
- III. Storing yearly values as separate columns makes analysis difficult.
- IV. Country, region and indicator information may be repeated.
- V. Comparing climate measurements across countries and years is inconvenient.
- VI. Large datasets require structured and optimized storage.



Objective

- I. Design a normalized relational database for climate data.
- II. Store country, region and income-group information.
- III. Store climate indicators and their metadata.
- IV. Convert yearly observations into structured measurement records.
- V. Provide advanced SQL-based climate analysis.
- VI. Enforce database rules using procedures and triggers.
- VII. Improve query performance using indexing.

TAE 1 Schema

MAIN ENTITIES

REGION

INCOME_GROUP

COUNTRY

LENDING_TYPE

CLIMATE_I
NDICATOR

CLIMATE_MEASUREMENT

INDICATOR_
METADATA

SOURCE_ORGANIZATION

KEY RELATIONSHIPS

REGION + COUNTRY

→ Country belongs to a region

INCOME_GROUP + COUNTRY

→ Country belongs to an income group

COUNTRY + CLIMATE_MEASUREMENT

→ Country has yearly climate measurements

CLIMATE_INDICATOR + CLIMATE_MEASUREMENT

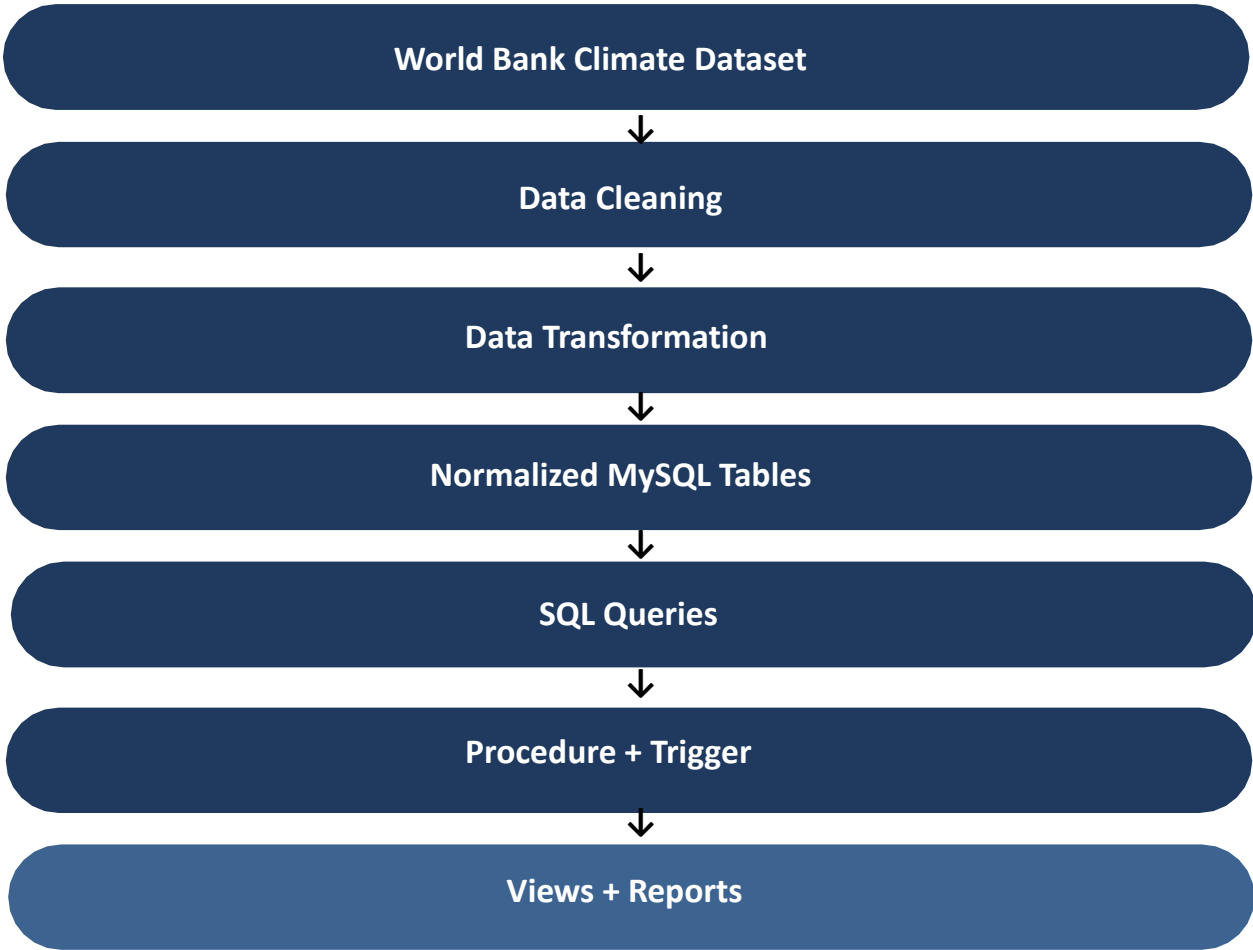
→ Indicator records climate measurements

Database Architecture & Setup

TECHNOLOGY STACK

Component	Technology
RDBMS	MySQL WorkBench
Query Language	SQL
Procedural Language	MySQL Stored Procedure
Development Environment	VS Code
Source Data	World Bank Climate Indicators Dataset
Data Generation	Python / Pandas

SQL IMPLEMENTATION PIPELINE



Database Implementation

DDL

- CREATE TABLE
- Primary Keys
- Foreign Keys
- NOT NULL
- UNIQUE
- CHECK constraints

Source Data

- I. World Bank climate dataset
- II. Country metadata
- III. Indicator metadata
- IV. Year-wise climate measurements
- V. Cleaned and normalized CSV files

DATA POPULATION

Table	Records
REGION	7
INCOME_GROUP	4
COUNTRY	217
CLIMATE_INDICATOR	51
INDICATOR_METADATA	28
SOURCE_ORGANIZATION	15
CLIMATE_MEASUREMENT	5248

Advanced SQL Queries

IMPLEMENTED SQL CONCEPTS

- Inner Join
- Left Outer Join
- Right Outer Join
- Self Join
- Aggregate Functions
- GROUP BY
- HAVING
- Subqueries
- Correlated Subqueries
- EXISTS / NOT EXISTS

EXAMPLE — CORRELATED SUBQUERY

```
SELECT
    c.country_name,
    ci.indicator_name,
    cm.measurement_year,
    cm.measurement_value
FROM climate_measurement cm
JOIN country c
    ON cm.country_id = c.country_id
JOIN climate_indicator ci
    ON cm.indicator_id =
    ci.indicator_id
WHERE cm.measurement_value >
(
    SELECT AVG(cm2.measurement_value)
    FROM climate_measurement cm2
    WHERE cm2.indicator_id =
    cm.indicator_id
);
```

Purpose: Find climate measurements whose value is higher than the average value of the same indicator.

Stored Procedure — Climate Measurement

PROCEDURE

```
add_climate_measurement()
```

PARAMETERS

```
country_id
```

```
indicator_id
```

```
measurement_year
```

```
measurement_value
```

EXAMPLE CALL

```
CALL add_climate_measurement(  
  1,  
  1,  
  2025,  
  45.678900 ););
```

PROCEDURE LOGIC

Check Measurement Year



Check Measurement Value



Validate Parameters



Insert Climate Measurement



Purpose

Automates insertion of climate measurements while validating the measurement data.

Trigger — Climate Measurement Audit

TRIGGER

`trg_measurement_update`

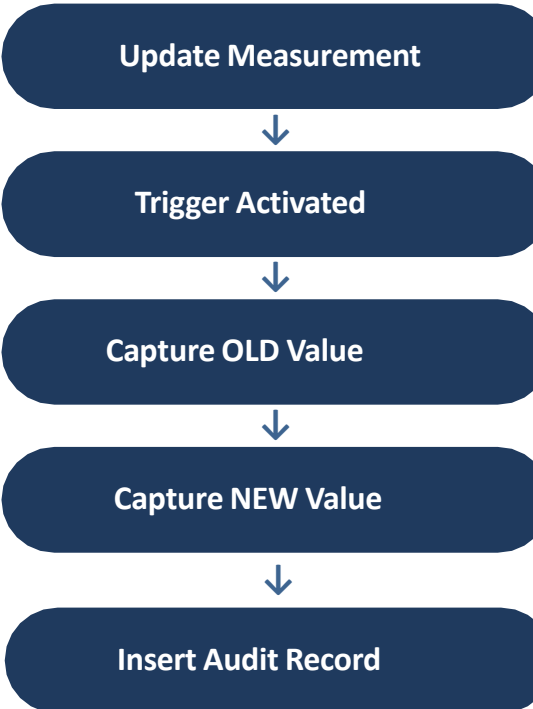
BUSINESS RULE

Whenever a climate measurement is updated, the previous and new values should be recorded in an audit table.

IMPLEMENTATION

```
CREATE TRIGGER trg_measurement_update
AFTER UPDATE ON climate_measurement
FOR EACH ROW
INSERT INTO climate_measurement_audit
VALUES (
    OLD.measurement_id,
    'UPDATE',
    OLD.measurement_value,
    NEW.measurement_value );
```

TRIGGER FLOW



Why it matters

Database changes are automatically tracked without depending on application-side logging..

Views & Climate Reporting

country_climate_summary

Provides:

- Country
- Region
- Climate Indicator
- Number of measurements
- Average value
- Minimum value
- Maximum value

regional_climate_summary

Provides:

- Region
- Climate Indicator
- Measurement year
- Measurement count
- Average value
- Minimum value
- Maximum value

Why Views?

- I. Pre-join complex climate data for reuse
- II. Simplify reporting queries
- III. Centralize analytical logic
- IV. Provide summarized climate information

EXAMPLE QUERY

```
CREATE VIEW country_climate_summary AS
SELECT
    c.country_name,
    AVG(cm.measurement_value) AS average_value
FROM country c
JOIN climate_measurement cm
    ON c.country_id = cm.country_id
GROUP BY c.country_name;
```

Conclusion

ACHIEVEMENTS

- ✓ Implemented the TAE-1 relational schema in MySQL.
- ✓ Populated the database with cleaned climate data.
- ✓ Converted yearly climate observations into normalized records.
- ✓ Implemented advanced SQL queries.
- ✓ Developed a parameterized stored procedure.
- ✓ Implemented an automated audit trigger.
- ✓ Created climate reporting views.
- ✓ Implemented indexes and composite indexes

FINAL OUTCOME

A structured MySQL database for global climate-change impact analysis and sustainability monitoring.

FUTURE SCOPE

Integration with continuously updated climate datasets

Real-time climate data updates

Web-based climate monitoring dashboard

Predictive climate analytics